

Dietary intervention versus metformin to improve the reproductive outcome in women with polycystic ovary syndrome. A prospective comparative study.

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OBJECTIVE: To compare the clinical results and reproductive outcome in obese women with polycystic ovary syndrome (PCOS) following dietary intervention or treatment with metformin. METHODS: Forty-six patients with PCOS were studied prospectively in Prince Rashed Hospital, Irbid, Jordan, between January 2003 and April 2005. The women were randomly divided into 2 groups: Group 1 (n=24) was prescribed with 1200-1400 kcal/day diet (25% proteins, 25% fat, and 50% carbohydrates plus 25-30 gm of fiber per week). Group 2 (n=22) was assigned to take 850 mg of metformin twice in a continuous manner. Both treatments continued for 6 months. Clinical and biochemical data, before and after both treatments along with the reproductive outcome were compared between the 2 groups. RESULTS: There were no significant differences between the 2 groups in terms of age, body mass index (BMI) and duration of infertility. Both groups had a significant improvement after treatment in the menstrual cyclicity (66.7% and 68.2% versus 12.5% and 18.2%) and significant reduction in BMI (mean of 27.4 and 27.8 versus 32.2 and 31.9), luteinizing hormone levels (7.9+/-1.7 and 6.9+/-1.8 versus 11.8+/-2.2 and 11.5+/-1.8), and androgen (testosterone, androstenedione, dehydroepiandrosterone sulfate) concentration. The clinical, biochemical, and reproductive outcome including menstrual cycle pattern, ovulation, and pregnancy rates were similar in both groups after treatment. CONCLUSION: Amelioration of hyperinsulinemia and hyperandrogenemia with dietary intervention or metformin treatment improves significantly the clinical features and reproductive function in overweight PCOS women.